**International Baccalaureate Diploma Programme**

**Extended Essay**

**Title:** The Effect of Citrus Fruits on Human Enamel Demineralisation: An Investigation Using Pig Teeth as a Model

**Research question:** To what extent do citrus fruits contribute to the enamel demineralization of pig teeth?

**Subject:** Biology

**Word count:** 3400

## Table of Contents

# Introduction

In this essay, I would like to explore how the acidic foods we eat affect our oral health, specifically teeth. Enamel is the primary protective barrier against chemical and physical damage of our teeth. However, despite the enamel's hardness, it is still vulnerable to demineralization from food acidity, a process of acids dissolving the mineral components of the enamel, weakening its structure over time. I found this interesting as this process is becoming more and more relevant in society, where dental health is becoming more and more important, and our diets often contain acidic beverages (soda) or foods (fruits). Fruits are an important part of diets for their nutritional contents, but they're also the most common sources of dietary acids that affect our teeth.

Enamel demineralization caused by citrus consumption has been a subject of scientific research, largely due to the chemical properties of citric acid ($C_6H_8O_7$), which lowers the pH and interacts with calcium in enamel. Few studies have focused specifically on natural acids in fruits and their erosive properties. Examining this connection gives us a better understanding of how everyday diet choices can influence our dental health. My results could also suggest a connection between natural chemistry and human concerns.

This essay investigates the research question: To what extent do citrus fruits contribute to enamel demineralization of pig teeth? Pig teeth are used here as a model system because of their structural and chemical similarities to human teeth/enamel, as pigs have the closest diets to humans do (Lopes et al., 2006) This makes them a reliable substitute for exploring the effects of acidic exposure in a controlled environment. The research also considers varying acidity levels of different citrus fruits, showing how these factors can correlate with the degree of enamel corrosion observed.

## Background information

**Composition of tooth enamel and corrosion effects on tooth enamel**

Tooth enamel is composed mainly of hydroxyapatite ( $Ca_{10}PO_{4}6OH_{2}$ ), a calcium phosphate that provides enamel to teeth with its exceptional hardness and resistance.

However, corrosion is one of the enamel vulnerabilities, as enamel is chemically vulnerable to acids because hydrogen ions from acids react with its hydroxide and phosphate ions, which destroy the lattice structure, causing calcium ions to be lost (Shellis et al., 2023). This process is known as demineralization, which weakens enamel and leads to erosion and tooth decay, caused by the acidity coming from foods and drinks (Hughes et al., 2000).

Demineralization happens in the mouth when the pH drops below an average value of around 5.5 (Shellis et al.2010). At this value, there is an equilibrium between enamel corrosion and the remineralization coming from the saliva of our mouth (Enax et al., 2024). Normally, our saliva also acts as a buffer to neutralize acids while also supplying calcium and phosphate ions to restore minerals lost in the enamel. However, when the enamel's exposure to acid, for example, is too frequent or long, such as from repeated consumption of citrus fruits, the saliva's buffering abilities can become overwhelmed, therefore leading to corrosion dominating in our mouths, leading to demineralization.

**Citrus acids and the reaction with enamel**

Citrus fruits such as lemons and limes are rich in this weak organic acid, with molecular formula *$C_6H_8O_7$*. Interestingly, each of its three carboxylic groups can dissociate and release hydrogen ions and contribute to a reduction in pH. The acid dissociation is represented by the equilibrium below:

$$H_3C_6H_5O_7 \rightleftharpoons H^+ + H_2C_6H_5O_7^-$$

Citric acid can donate multiple protons under acidic conditions. In addition to its acidity, citric acid also has chelating properties, meaning it can bind to calcium ions to form soluble calcium citrate complexes (Zhu et al., 2013). These two effects not only lower pH but also remove calcium, making citric acids very effective at causing enamel corrosion.

While there have been many studies on processed acids (such as phosphoric acid in soda drinks), fewer researchers have focused on naturally occurring fruit acids and their negative effects on tooth enamel. Fruit juice differs in species and ripeness; hence, the pH varies. For example, lemon juice typically has a pH of roughly 1.0, whereas limes fall somewhere around 2.0 on the scale. These pH differences indicate that the fruits are not equally responsible for tooth enamel corrosion. The differences in these acids can have big implications on the demineralization of our teeth, which is why it is important to understand how all that affects our dental health.

Although many studies have focused on industrially produced acids such as phosphoric acid in soda drinks, not many have researched naturally produced acids in fruits and their effects on enamel. The chemical composition of fruit juices varies depending on species and ripeness, resulting in different pH levels. For instance, lemon juice normally has a pH of around 2.0, while limes have a pH of around 3.0. These differences in pH suggest that the fruits don't contribute equally to the corrosion of enamel. Therefore, understanding how each acid's strength and composition affects the demineralization of our teeth is important to evaluate their impact on our dental health.

Looking from a chemical standpoint, enamel dissolution/corrosion can be expressed by the reaction:

$$Ca_{10}(PO_4)_6(OH)_2 + 8H^+ \rightarrow 10Ca^{2+} + 6HPO_4^{2-} + 2H_2O$$

This reaction shows how an increase in hydrogen ion concentration causes the process toward demineralization, which is consistent with Le Chatelier's Principle. As the acidity of the surrounding solution increases, the equilibrium shifts rightward, favoring the products, including mineral loss and calcium ion release. Moreover, when pH rises, the equilibrium shifts leftward toward reactants, allowing for remineralization. Hence, differences in fruit acidity and buffering capacity of various fruits can produce visible differences in the degree of enamel demineralization.

**Correlation between pig teeth and human teeth**

In science experiments, researchers often use pig teeth as a common substitute for human teeth due to ethical reasons. Pig enamel shares similar mineral content, structure, and hardness of human enamel, making pig teeth a reliable and comparative model for studies (Lopes et al.2006). By exposing pig enamel samples to different citrus fruit juices and measuring chemical and physical changes, such as enamel mass loss and structure changes, one can approximate the effects that these fruits would have on human enamel.

This study uses basic chemistry to connect how strong an acid is to visible changes in tooth enamel. By looking at the reactions, it aims to explore how citrus fruits cause enamel to wear down and how natural acids in our diet affect our teeth in daily life.

## Hypothesis

I predict that there will be a high degree of enamel demineralization, especially with the citrus fruits with higher citric acid concentrations and lower pH values. Specifically, lemon juice is expected to cause the most significant loss of enamel minerals compared to lime. I predict this because, based on the chemical principle that increasing the hydrogen ion concentration will shift the equilibrium of the dissolution reaction toward the products, the release of calcium and phosphate ions will occur. Also, the chelating properties of citric acids, as mentioned, can bind to calcium ions, enhancing the acid's ability to demineralize the enamel of the pig teeth. Therefore, the overall extent of enamel corrosion will be directly correlated to the acidity of each citrus fruit juice.

# Method

Independent variables

- Type of citrus fruit juice used (lemon, lime).

Dependent variables

- Change in pH of the juice before and after contact with the teeth samples.
- Mass change of the enamel sample before and after immersion to verify mineral loss

Controlled variables

- Volume of juice used for each trial (25ml)
- Surface area and size of each enamel sample.
- Exposure duration of juice with teeth (7 days)
- Temperature (kept constant around 20 degrees in a lab)
- Source and preparation of enamel (same type of pig teeth, disinfected and dried uniformly).
- pH probe calibration
- Time between the extraction of juice and the exposure of the teeth (to prevent pH change).

# Risk Assessment and Safety

Bacterial growth:

There is a risk that pig teeth are unclean and can react with fruit juices to create bacteria within 7 days, which can spread into fungus. To prevent this, the temperature was set to 4°C in a fridge rather than at normal room temperature in an open environment. Colder temperature prevents bacteria and fungi from growing rapidly, which can create potential health risks for people and students inside the school. To make this experiment safe, the pig teeth were disinfected, and the experiment was done in closed environments.

## Materials

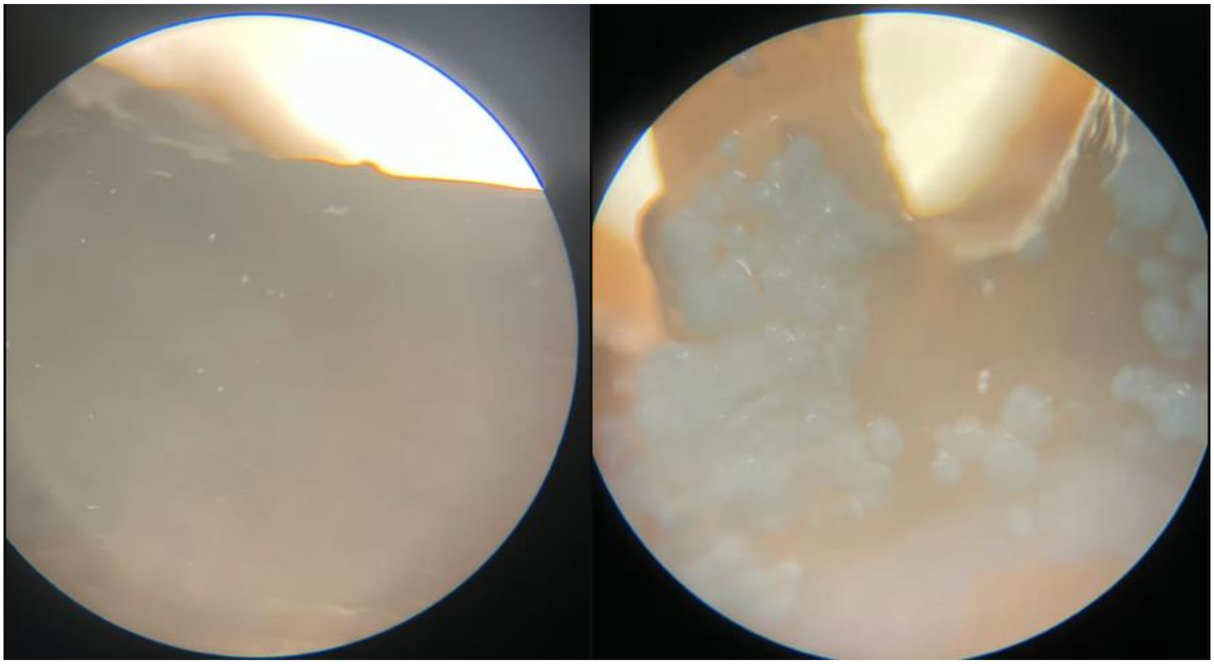
1. Sterilized Beakers, a knife and 20 test tubes
2. 100ml of fresh lemon and lime juice
3. Permanent marker
4. Clear tape for covering test tubes
5. Filter paper
6. 20 Pig Incisor teeth

## Developed method

The experiment method was developed to make sure it was a fair comparison between the effects of the two different citrus fruit juices on enamel demineralization. The choice of fruit type was based on their differences in citric acid content and pH, providing a wider range of acidity without using any other unnatural acid solutions. I chose to measure mass loss for the most direct approach to finding the degree of mineral loss, allowing me to do physical comparisons and mathematical calculations.

**Figure 1:**

*Before and after photos of the incisors in fruit juice for 7 days*



According to Figure 1, the final enamel is different from the start, with only a few spots of enamel left intact on the teeth compared to the first photo. This clearly indicates that the fruit juice has eroded the teeth and shows that the level of acid does affect the level of enamel demineralization.

After one failed experiment where the environments weren't closed, and bacteria grew on the juices, I improved my method. After sterilizing the pig molars and drying them with paper towels, I weighed the starting mass of each tooth and labeled them while placing them into each corresponding flask of fruit juice and noting down the starting mass and pH of everything. Finally, I moved all flasks into the lab to maintain a constant environment and left them untouched in the lab for 7 days and checked them after a week. This finalized procedure gives a controlled and justified design to research how the acidity of citrus fruits influences enamel demineralization, supporting the overall aim of the experiment.

I chose to use pig teeth as a model for human enamel since they have a close similarity in their composition (Lopes et al.2006). To minimize differences, I chose and cleaned 20 incisors of the pig teeth of approximately equal size. Then I squeezed fresh fruit juices into beakers to make sure the acidity hadn't oxidized, and the solutions were filtered to remove pulp, which could interfere with enamel contact, while also measuring the pH of the fruit juices using a digital pH meter calibrated using buffer solutions at pH 4.0 and 7.0. Then, all samples were also to be immersed at the same time in the same conditions in a fridge to minimize environmental differences.

## Results

After 7 days have passed, I analyze the changes in the result by measuring the changes in the mass of pig teeth after exposure to lime and lemon juice. The pH of lime juice was 1.9, while lemon juice had a lower pH of 1.54. For each condition, the starting and final mass of ten pig teeth was measured using an electronic balance.

Below is the raw data collected in table form:

| teeth | mass original (grams) | mass final (grams) |
|---|---|---|
| lime 1 | 6.4 | 6.3 |
| lime 2 | 4.4 | 3.9 |
| lime 3 | 6.2 | 5.6 |
| lime 4 | 6.2 | 5.8 |
| lime 5 | 6.0 | 5.8 |
| lime 6 | 7.0 | 6.4 |
| lime 7 | 5.8 | 5.3 |
| lime 8 | 5.5 | 5.3 |
| lime 9 | 6.7 | 6.5 |
| lime 10 | 6.4 | 6.2 |
| lemon 1 | 4.6 | 4.2 |
| lemon 2 | 4.9 | 3.4 |
| lemon 3 | 5.4 | 4.8 |
| lemon 4 | 4.4 | 4.0 |
| lemon 5 | 5.8 | 5.0 |
| lemon 6 | 3.6 | 3.4 |
| lemon 7 | 5.6 | 5.3 |
| lemon 8 | 3.5 | 3.0 |
| lemon 9 | 4.4 | 4.2 |
| lemon 10 | 5.1 | 4.8 |
| lime mean | 6.06 | 5.71 |
| lemon mean | 4.73 | 4.21 |

For the lime juice trials, every tooth showed a decrease in mass after the 7-day exposure period. The mean initial mass of all ten teeth was 6.06g, while the mean final mass was 5.71g, resulting in an average mass loss of 0.35g. Individual mass losses ranged from 0.1g to 0.6g, showing different variations between the same samples but a similar overall reduction in mass across all ten teeth.

For the lemon juice trials, a decrease in mass can also be observed in all ten teeth after 7 days. The mean initial mass was 4.73g, and the mean final mass was 4.21g, giving an average of 0.52g lost. Individual mass loss ranged from 0.1g to 1.5g. Overall, the lemon-soaked teeth group showed much greater mass loss than the lime-treated group.

## Processed data

**Table 1**

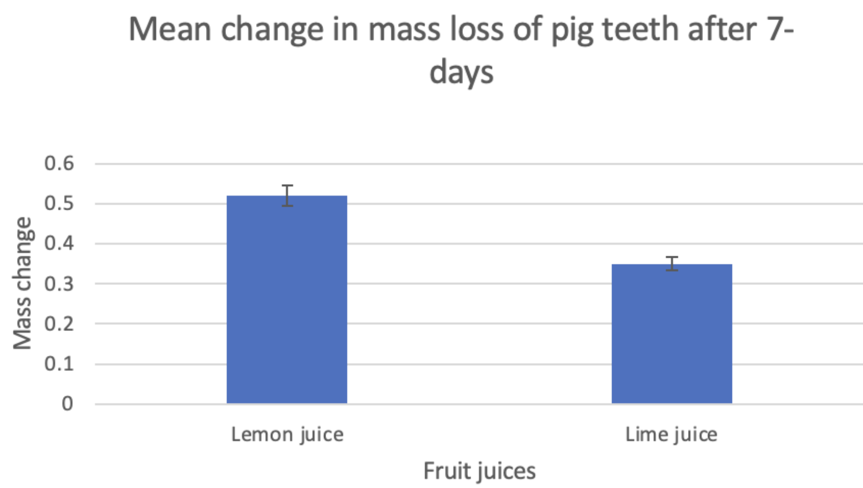
*Mass changes of the individual teeth:*

| | trial 1 mass change | trial 2 | trial 3 | trial 4 | trial 5 | trial 6 | trial 7 | trial 8 | trial 9 | trial 10 | Mean | Standard DV |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lemon juice | 0.4 | 1.5 | 0.6 | 0.4 | 0.8 | 0.2 | 0.3 | 0.5 | 0.2 | 0.3 | 0.52 | 0.39 |
| Lime juice | 0.1 | 0.5 | 0.6 | 0.4 | 0.2 | 0.6 | 0.5 | 0.2 | 0.2 | 0.2 | 0.35 | 0.18 |

This table shows the individual tooth mass changes for both lime and lemon juices. It is seen that the teeth soaked in lemon juice for 7-days has a more significant mean mass lost then when compared to the lime juice. This follows my hypothesis that higher acidity will result in a higher mass loss in enamel due to erosion. To see the correlation more clearly i will use this data to do a T-test to see if the results of mass loss are dependent or independent of the acidity of the juices.

**Table 1**

*Mean change in mass loss of pig teeth after 7-days*




A decrease in mass was observed in all ten teeth. The extent of mass loss varied between samples, with some teeth having much larger reductions than others, though the overall trend was a decrease in mass across all teeth. Figure 2 shows the comparison between the mean percentage mass lost from pig teeth to lime and lemon juice. Teeth exposed to lemon juice showed a higher mean percentage mass loss than those in lime juice.

**Figure 2**

*T-test calculations on the GDC calculator*

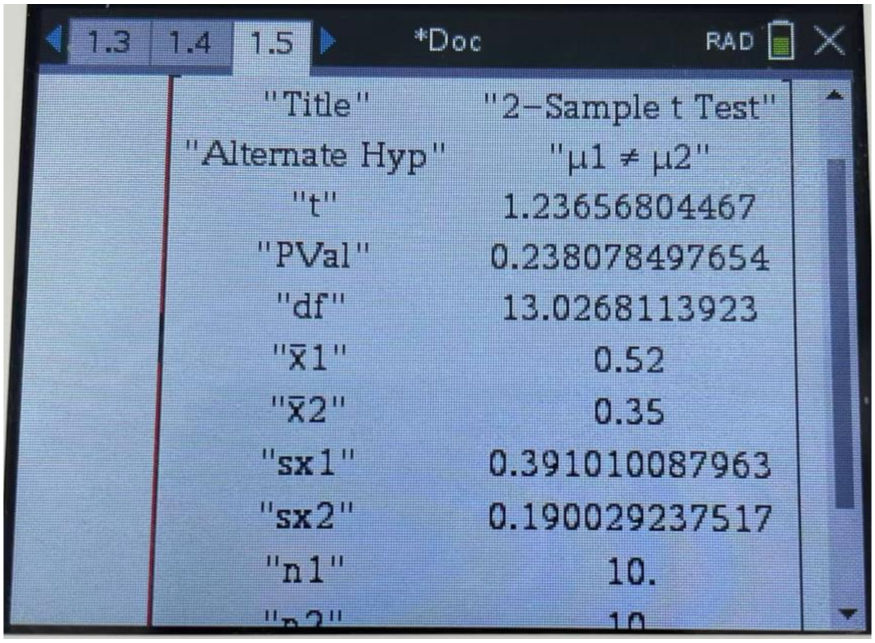



Figure 2 above is a screenshot of the T-test done using chart 1 of the mass changes. A two-sample t-test was done to compare the mean mass loss between lemon and lime-soaked teeth. The p-value obtained was 0.238, which is a lot bigger than 0.05. Therefore, the difference in mean mass loss between the two groups is not statistically significant at a 95% confidence level.

According to the T-test, lemon had a higher mean loss, but the p-value shows 0.238, meaning there's a 23.8% chance that the difference happened randomly, which is too high to confidently say lemon truly does more demineralization over time than lime. This doesn't mean my hypothesis is wrong; it just means my data isn't strong enough to support my hypothesis. Therefore, the null hypothesis is not rejected at the 95% confidence level. However, data show that the trend aligns with the expected chemical principles.

## Discussion

Table 1 showed that the teeth exposed to lemon juice experienced a greater mass loss than the teeth in lime juice. However, the calculated p-value of 0238 shows that this difference is not statistically significant at a 95% confidence level. Meaning that the difference in the two groups could be natural variability rather than a difference in erosive potential. Because mass loss was used as the main measure of demineralization, calculating the results as a percentage mass loss would provide a more accurate comparison between samples of different sizes, since larger teeth contain more enamel mass. Using percentage loss strengthens the validity of comparisons by removing variation in starting mass and improving the reliability of interpretation.

Despite the P-value from the calculations, the overall pattern is consistent with my hypothesis. Lemon juice with the lower pH has a higher hydrogen ion concentration than lime juice; the greater mean mass loss seen in lemon-treated samples aligns with expectations based on the hydroxyapatite equilibrium and Le Chatelier's principle. However, even though data suggests a directional relationship between acidity and enamel demineralization, the strength of the evidence isn't strong as the T-test proves that there is no certain correlation.

Looking at the results in a real-life context, this investigation changed the way I think about everyday dietary choices. Fruits like lemons and limes are thought of as healthy; this experiment shows that their acidity can have measurable effects on enamel when exposure is prolonged. While also showing how relatively small differences in pH can translate into noticeable physical changes over time. While in real life, teeth are not soaked in the juice for seven days, but repeated exposure to acidic foods and drinks can gradually create similar effects, highlighting the importance of why dental professionals recommend rinsing and cleaning after consuming high-acidity products. This investigation helped me gain a clearer understanding of how small chemical principles directly affect long term oral health

## Evaluation

The results collected in this experiment are consistant to my expectation outlined in the introduction that lower pH solutions cause greater enamel demineralization. After seven days of exposure, pig teeth in the lemon juice showed a higher mean mass loss than the teeth in lime juice. This supports the predicted connection between acidity and enamel erosion in acidic conditions.

As discussed in the introduction, enamel is made of hydroxyapatite and dissolves in acidic environments. The greater mass loss weighted from teeth in lemon juice can be attributed to its lower pH (1.54) compared to lime juice (1.90), resulting in a higher hydrogen ion concentration across the 7-day duration.

A small difference in pH can result in a large change in hydrogen ion concentration. By using the hydrogen ion concentration equation, we can look at the two juices individually to see the difference:

$$pH = -\log_{10} [H^+]$$

For lime juice: [H+] =10^{-1.90}=0.01258925412

For lemon juice: [H+] =10^{-1.54} =0.02884031503

Comparing the two fruit juices, lemon juice (2.88) has more than double the hydrogen ion concentration of lime juice (1.44). Even though the actual difference in pH is relatively small, the chemical difference is huge, which would explain the greater enamel demineralization in the pig teeth. This acidic environment would increase the extent of hydroxyapatite dissolution, as shown by the larger mass reductions measured.

Uncertainties were a focus during this investigation. Because any uncertainties will affect the calculated mass loss directly. One source of uncertainty comes from the precision of equipment such as the balance used to measure the starting and final tooth mass. To minimize random error, each tooth was weighed three times, and an average value was used. However, systemic error can still be unpreventable if the balance was not perfectly calibrated.

Another source of uncertainty comes from the pH probe, which can be mis-calibrated. To ensure that the probe was accurate, two pH buffers of pH 4 and pH 6 were used to calibrate the probe. But the pH probe may still be imprecise because of systemic errors from the sensor of the probe.

The enamel samples also didn't have a controlled surface area. Bigger pig incisors have more surface area that can react with the fruit juice, leading to a greater mass loss. Along with the roots of the incisors that don't contain enamel, that are uncertain if they have lost mass during the erosion in the fruit juices, could contribute to the uncertainty of the experiment.

A limitation of this experiment is that pig teeth were continuously soaked in acidic solutions for 7 days, which doesn't represent real dietary exposure accurately, where the teeth's contact with acids is discontinuous with saliva that remineralizes teeth (Farooq and Bugshan 2020). In addition, pig teeth enamel isn't identical to human enamel, which may limit the generalization of the results.

Meanwhile, another limitation of the investigation is that the acidity of the fruit juices may have changed over the 7-day exposure period. Although the starting pH of each juice was measured, factors such as oxidation and evaporation could have altered the concentration of citric acid in the fruit juices. As a result, the teeth may not have been exposed to the same acidic environment throughout the experiment, which may have influenced the rate of enamel mass lost.

To improve the investigation, future experiments should include cycles of acid exposure so that it simulates real dietary erosion. Additionally, the exposed surface area of enamel should be standardized through coating parts of the tooth, which would reduce variability and improve the precision of mass change measurements. Finally, there were no calcium measurements. By measuring the calcium ion concentration, such as using a reflectometer (Carvalho et al.2022) would have directly looked at the hydroxyapatite loss and strengthened the explanation of enamel corrosion.

Despite this variability, the overall credibility of the results is supported by the consistent trend seen across all teeth trials. All teeth exposed to lime and lemon juice showed a reduction in mass, indicating a clear and repeatable pattern of enamel demineralization. By using ten teeth per acid, this increases the accuracy of the data, and by maintaining constant exposure, juice volume, and measurement procedures, this improves the validity of comparisons between the two acids.

## Conclusion

This investigation is aimed at looking at the effect of acidic fruit juices on enamel demineralization by measuring the mass change of pig teeth after seven days of acid exposure. Results showed that all teeth decreased in mass, which shows enamel loss, with teeth in lemon juice showing the greatest mean percentage mass loss compared to those in lime juice. As lemon juice has a lower pH than lime juice, the findings support the conclusion that increased acidity leads to greater enamel demineralization over prolonged exposure. Overall, the investigation has demonstrated a clear relationship between the pH of dietary acids and the extent of enamel erosion.

# Reference List

Carvalho, T. S., Baumann, T., & Lussi, A. (2016). A new hand-held optical reflectometer to measure enamel erosion: correlation with surface hardness and calcium release. *Scientific Reports*, *6*(1). Retrieved February 15, 2026 from https://doi.org/10.1038/srep25259

Enax, J., Fandrich, P., Schulze, E., & Epple, M. (2024). The Remineralization of Enamel from Saliva: A Chemical Perspective. *Dentistry Journal*, *12*(11), 339–339. Retrieved February 15, 2026 from https://doi.org/10.3390/dj12110339

Hughes, J. A., West, N. X., Parker, D. M., van den Braak, M. H., & Addy, M. (2000). Effects of pH and concentration of citric, malic and lactic acids on enamel, in vitro. *Journal of Dentistry*, *28*(2), 147–152. Retrieved February 15, 2026 from https://doi.org/10.1016/S0300-5712(99)00060-3

Lopes, F. M., Markarian, R. A., Sendyk, C. L., Duarte, C. P., & Arana-Chavez, V. E. (2006). Swine teeth as potential substitutes for in vitro studies in tooth adhesion: A SEM observation. *Archives of Oral Biology*, *51*(7), 548–551. Retrieved February 16, 2026 from https://doi.org/10.1016/j.archoralbio.2006.01.009

Shellis, R. P., Barbour, M. E., Jones, S. B., & Addy, M. (2010). Effects of pH and acid concentration on erosive dissolution of enamel, dentine, and compressed hydroxyapatite. *European Journal of Oral Sciences*, *118*(5), 475–482. Retrieved February 15, 2026 from https://doi.org/10.1111/j.1600-0722.2010.00763.x

Shellis, R. P., Barbour, M. E., Parker, D. M., Addy, M., & Lussi, A. (2023). Effects of Calcium and Phosphate on Dissolution of Enamel, Dentin and Hydroxyapatite in Citric Acid. *Swiss Dental Journal*, *133*(7/8), 432–438. Retrieved February 5, 2026 from https://doi.org/10.61872/sdj-2023-07-08-02